Adjacency List

Declaration with size n:

```
vector<vector<int>> adj(n);
```

Access vertex i's neighbors:

```
for (int neighbor : adj[i]) {
    // process neighbor
}
```

Access specific neighbor (j-th neighbor of vertex i):

```
int neighbor = adj[i][j];  // j-th neighbor in the list
```

Add edge (i → j):

```
adj[i].push_back(j);
```

Adjacency Matrix

Declaration with size n×n:

```
vector<vector<int>> matrix(n, vector<int>(n, 0));
```

Access vertex i's neighbors (loop through all vertices):

```
for (int j = 0; j < n; j++) {
    if (matrix[i][j] == 1) {  // if edge exists
        // j is a neighbor of i
    }
}
```

Check if j is neighbor of i:

```
if (matrix[i][j] == 1) {
    // yes, edge exists from i to j
```

```
}
```

Add edge (i → j):

```
matrix[i][j] = 1;
```

Quick Comparison

Operation	Adjacency List	Adjacency Matrix
Get all neighbors of vertex i	`for (int v : adj[i])`	`for (int j = 0; j < n; j++) if (matrix[i][j])`
Check if edge i→j exists	Loop through `adj[i]` (O(degree))	`matrix[i][j] == 1` (O(1)) [1] [2]
Space	O(V + E)	O(V²) [3]
Best for	Sparse graphs	Dense graphs, quick edge lookups [1] [3]

Key difference: adjacency list directly stores neighbors, matrix must scan all n vertices to find neighbors.[2] [4]

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1. https://www.geeksforgeeks.org/dsa/adjacency-matrix/
2. https://www.geeksforgeeks.org/dsa/implementation-of-bfs-using-adjacency-matrix/
3. https://stackoverflow.com/questions/2218322/what-is-better-adjacency-lists-or-adjacency-matrices-for-graph-problems-in-c
4. https://www.cipherschools.com/blogs/others/graphs-representation/
5. https://csiflabs.cs.ucdavis.edu/~tyfeng/ecs34/C++/graphs2.html
6. https://bridgesuncc.github.io/tutorials/Graph_AM.html
7. https://www.cs.fsu.edu/~lacher/lectures/Output/graphs1/index.html?%24%24%24slide05.html%24%24%24
8. https://www.geeksforgeeks.org/cpp/cpp-program-to-implement-adjacency-matrix/
9. https://cs.kenyon.edu/index.php/scmp-218-00-data-structures/adjacency-matrix-in-c/
10. https://www.shecodes.io/athena/9971-create-adjancency-matrix-with-c